



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For



CompTIA Cybersecurity Analyst (CySA+) Training

TGS Ref No: TGS-2024049211

Version 1.0

Document Version Control Record

Version	Effective Date	Summary of Changes	Author
1.0	6 July 2026	First edition — 5-day lesson plan aligned to the 30 hands-on CySA+ labs, with slide-number references to the Learner Guide Slides deck.	Tertiary Infotech Academy Pte Ltd

Course Details

Field	Detail
Course Title	CompTIA Cybersecurity Analyst (CySA+) Training
Course Code	TGS-2024049211
WSQ TSC	Cyber Risk Management (ICT-SNA-4007-1.1)
Duration	5 Days (40 hours)
Daily Schedule	9:00 AM – 6:00 PM (8 training hours/day, excluding lunch)
Lunch Break	12:30 PM – 1:15 PM (45 minutes)
Breaks	Short tea breaks are scheduled within each day's training hours
Delivery Mode	Instructor-led presentation, live demonstration and hands-on labs (Killercodea / free tools)
Assessment	Written Assessment (SAQ) 1 hr + Case Study (CS) 1 hr — Open Book (Day 5)
Prerequisites	Basic networking and Linux command-line familiarity; interest in cybersecurity operations
Slide Deck	CompTIA Cybersecurity Analyst (CySA+) Training-v6.pptx (172 slides)

Note: the 'Slides' column in each daily schedule references the exact slide numbers in the Learner Guide Slides deck for that activity.

Course Overview

This 5-day hands-on course develops practical cybersecurity analyst skills aligned to the CompTIA CySA+ CS0-003 exam. Across four domains — Security Operations, Vulnerability Management, Incident Response, and Reporting & Communication — learners complete 30 step-by-step labs using the free Killercodea playground and free industry tools, culminating in a written and case-study assessment.

Learning Outcomes

By the end of this course, participants will be able to:

- Identify cyber security risks, threats, and vulnerabilities, and assess their impact on the organization.
- Develop cyber risk assessments by consolidating insights from business to identify security gaps.
- Identify possible treatments for identified cyber risks, threats, and vulnerabilities.
- Implement endorsed treatment measures to address security gaps and ensure data protection.

Day 1

Topic 1: Security Operations

Morning Session · 9:00 AM – 12:30 PM

Time	Topic / Activity	Duration	Slides
9:00 – 9:20	Digital Attendance (AM); Trainer & Learner Introduction; Ground Rules	20 min	1–5
9:20 – 9:55	Lesson Plan; Learning Outcomes; Skills Framework; Course Outline	35 min	6–19
9:55 – 10:15	Final Assessment Briefing; Criteria for Funding; LMS/TMS	20 min	20–22
10:15 – 10:30	Tea Break	15 min	—
	Topic 1: Security Operations — Topic Overview	—	23–24
10:30 – 12:30	Lab 1: Log Ingestion and Time Synchronization; Lab 2: OS Hardening and System Process Inspection; Lab 3: Network Segmentation and Zero Trust; Lab 4: Detecting Malicious Network Activity; Lab 5: Host-Based IOC Hunting	120 min	25–47
12:30 – 1:15	Lunch Break	45 min	—

Afternoon Session · 1:15 PM – 6:00 PM

Time	Topic / Activity	Duration	Slides
1:15 – 1:20	Digital Attendance (PM)	5 min	—
1:20 – 3:30	Lab 6: Packet Capture for Threat Hunting; Lab 7: SIEM Log Correlation; Lab 8: Email Header Analysis (SPF/DKIM/DMARC)	130 min	48–60
3:30 – 3:45	Tea Break	15 min	—
3:45 – 6:00	Lab 9: Malware Triage with Hashing and VirusTotal; Lab 10: Threat Intelligence and MITRE ATT&CK	135 min	61–70
6:00	End of Day 1	—	—

Day 2

Topic 2: Vulnerability Management

Morning Session · 9:00 AM – 12:30 PM

Time	Topic / Activity	Duration	Slides
9:00 – 9:10	Digital Attendance (AM); Recap of previous day	10 min	—
	Topic 2: Vulnerability Management — Topic Overview	—	71–72
9:10 – 10:30	Lab 11: Asset Discovery with Nmap; Lab 12: Vulnerability Scanning with OpenVAS (Greenbone CE); Lab 13: Web App Scanning with OWASP ZAP	80 min	73–86
10:30 – 10:45	Tea Break	15 min	—
10:45 – 12:30	Lab 14: Web Recon with Nikto and Burp Suite; Lab 15: Metasploit Framework Basics	105 min	87–95
12:30 – 1:15	Lunch Break	45 min	—

Afternoon Session · 1:15 PM – 6:00 PM

Time	Topic / Activity	Duration	Slides
1:15 – 1:20	Digital Attendance (PM)	5 min	—
1:20 – 3:30	Lab 16: CVSS Scoring and Prioritization; Lab 17: Exploiting and Mitigating XSS and SQL Injection	130 min	96–104
3:30 – 3:45	Tea Break	15 min	—
3:45 – 6:00	Lab 18: Patch Management and Hardening; Lab 19: Attack Surface Reconnaissance	135 min	105–112
6:00	End of Day 2	—	—

Day 3

Topic 3: Incident Response and Management

Morning Session · 9:00 AM – 12:30 PM

Time	Topic / Activity	Duration	Slides
9:00 – 9:10	Digital Attendance (AM); Recap of previous day	10 min	—
	Topic 3: Incident Response and Management — Topic Overview	—	113–114
9:10 – 10:30	Lab 20: Cyber Kill Chain and ATT&CK Mapping; Lab 21: Evidence Acquisition and Chain of Custody	80 min	115–124
10:30 – 10:45	<i>Tea Break</i>	15 min	—
10:45 – 12:30	Lab 22: Memory Forensics with Volatility	105 min	125–129
12:30 – 1:15	<i>Lunch Break</i>	45 min	—

Afternoon Session · 1:15 PM – 6:00 PM

Time	Topic / Activity	Duration	Slides
1:15 – 1:20	Digital Attendance (PM)	5 min	—
1:20 – 3:30	Lab 23: Log Analysis for Incident Response; Lab 24: Containment with Host Isolation	130 min	130–138
3:30 – 3:45	<i>Tea Break</i>	15 min	—
3:45 – 6:00	Lab 25: Incident Response Playbook and Tabletop	135 min	139–143
6:00	End of Day 3	—	—

Day 4

Topic 4: Reporting and Communication

Morning Session · 9:00 AM – 12:30 PM

Time	Topic / Activity	Duration	Slides
9:00 – 9:10	Digital Attendance (AM); Recap of previous day	10 min	—
	Topic 4: Reporting and Communication — Topic Overview	—	144–145
9:10 – 10:30	Lab 26: Vulnerability Management Report; Lab 27: Executive Incident Report	80 min	146–152
10:30 – 10:45	<i>Tea Break</i>	15 min	—
10:45 – 12:30	Lab 28: Security Metrics Dashboard (MTTD / MTTR)	105 min	153–156
12:30 – 1:15	<i>Lunch Break</i>	45 min	—

Afternoon Session · 1:15 PM – 6:00 PM

Time	Topic / Activity	Duration	Slides
1:15 – 1:20	Digital Attendance (PM)	5 min	—
1:20 – 3:30	Lab 29: Compliance Reporting (PCI DSS / ISO 27001 / NIST CSF)	130 min	157–160
3:30 – 3:45	<i>Tea Break</i>	15 min	—
3:45 – 6:00	Lab 30: Stakeholder Communication and Lessons Learned	135 min	161–164
6:00	End of Day 4	—	—

Day 5

Revision and Final Assessment

Morning Session · 9:00 AM – 12:30 PM

Time	Topic / Activity	Duration	Slides
9:00 – 9:10	Digital Attendance (AM)	10 min	—
9:10 – 10:30	Revision: Topic 1 Security Operations & Topic 2 Vulnerability Management	80 min	25–70, 73–112
10:30 – 10:45	Tea Break	15 min	—
10:45 – 12:30	Revision: Topic 3 Incident Response & Topic 4 Reporting; Practice Exams	105 min	115–143, 146–164, 165
12:30 – 1:15	Lunch Break	45 min	—

Afternoon Session · 1:15 PM – 6:00 PM

Time	Topic / Activity	Duration	Slides
1:15 – 1:20	Digital Attendance (PM)	5 min	—
1:20 – 3:30	Q&A; Summary; exam techniques; Course Feedback & TRAQOM Survey	130 min	166–168
3:30 – 3:45	Tea Break	15 min	—
3:45 – 4:00	Assessment Briefing; Digital Attendance (Assessment)	15 min	169
4:00 – 6:00	Final Assessment: Written (SAQ) 1 hr + Case Study (CS) 1 hr	120 min	—
6:00	End of Class; Recommended Courses; Support	—	170–172

Slide Reference Summary

Section	Content	Slides
Front Admin	Title, Attendance, Ground Rules, Lesson Plan, Skills Framework, Learning Outcomes, Course Outline, Assessment, Funding, LMS	1–22
Topic 1: Security Operations	Section + Overview + Labs 1–10	25–70
Topic 2: Vulnerability Management	Section + Overview + Labs 11–19	73–112
Topic 3: Incident Response and Management	Section + Overview + Labs 20–25	115–143
Topic 4: Reporting and Communication	Section + Overview + Labs 26–30	146–164
Lab 1	Log Ingestion and Time Synchronization	25–29
Lab 2	OS Hardening and System Process Inspection	30–33
Lab 3	Network Segmentation and Zero Trust	34–38
Lab 4	Detecting Malicious Network Activity	39–42
Lab 5	Host-Based IOC Hunting	43–47
Lab 6	Packet Capture for Threat Hunting	48–52
Lab 7	SIEM Log Correlation	53–56
Lab 8	Email Header Analysis (SPF/DKIM/DMARC)	57–60
Lab 9	Malware Triage with Hashing and VirusTotal	61–65
Lab 10	Threat Intelligence and MITRE ATT&CK	66–70
Lab 11	Asset Discovery with Nmap	73–77
Lab 12	Vulnerability Scanning with OpenVAS (Greenbone CE)	78–82
Lab 13	Web App Scanning with OWASP ZAP	83–86
Lab 14	Web Recon with Nikto and Burp Suite	87–90
Lab 15	Metasploit Framework Basics	91–95
Lab 16	CVSS Scoring and Prioritization	96–99
Lab 17	Exploiting and Mitigating XSS and SQL Injection	100–104
Lab 18	Patch Management and Hardening	105–108
Lab 19	Attack Surface Reconnaissance	109–112
Lab 20	Cyber Kill Chain and ATT&CK Mapping	115–119
Lab 21	Evidence Acquisition and Chain of Custody	120–124
Lab 22	Memory Forensics with Volatility	125–129
Lab 23	Log Analysis for Incident Response	130–134
Lab 24	Containment with Host Isolation	135–138
Lab 25	Incident Response Playbook and Tabletop	139–143
Lab 26	Vulnerability Management Report	146–149
Lab 27	Executive Incident Report	150–152
Lab 28	Security Metrics Dashboard (MTTD / MTTR)	153–156
Lab 29	Compliance Reporting (PCI DSS / ISO 27001 / NIST CSF)	157–160
Lab 30	Stakeholder Communication and Lessons Learned	161–164
Back Admin	Practice Exams, Summary/Q&A, TRAQOM, Attendance, Final Assessment, Recommended Courses, Support, Thank You	165–172